

and the epoxy package is designed as an IR filter.

When a short IR burst is received by IRX1 (as you wave your hand in front of the switch), the demodulated pulses are fed to the trigger input (pin 2) of the second LMC555 (IC2). This, in turn, triggers the monostable wired around IC2 and its output pin 3 goes high for a period determined by the 2.2-mega-ohm potentiometer and capacitor C5. This turns off the standby indicator (LED1) and transistor T1 conducts to drive the 5V relay (RL1). LED1 enables you to locate the switch in the dark. AC mains supply to the load to be switched-on is routed through the pole and normally-opened contacts of RL1 as shown in the diagram. The circuit works off regulated 5V DC.

Fig. 2 shows the pin configurations of TSOP1738, IR LED1 and transistor BC547. Assemble the circuit on a

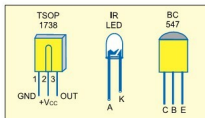


Fig. 2: Pin configurations of TSOP1738, IR LED and BC547

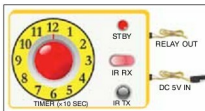


Fig. 3: Suggested enclosure

general-purpose PCB and enclose in a small plastic cabinet. Fit IR LED1 with a reflecting hood at a recessed position on the front panel of the enclosure. The dome-shaped face of the TSOP1738

should stick out from the front panel. Fit the time-control potentiometer (VR2) in an appropriate position. Finally, fit the standby indicator LED1 inside a suitable LED holder such that it slightly protrudes from the front panel. To prevent unwanted reflection of the IR beam, the finished unit should be mounted such that it does not face a nearby wall.

Using high-precision linear potentiometer VR2 and capacitor C5 (100µF), the time length can be set from nearly 1 second to 120 seconds. Attach a small paper dial on the front panel of the enclosure and mark various positions of the control knob of VR2 as shown in Fig. 3. The accuracy of the timer depends mainly upon the quality (and value) of timing capacitor C5. In practice, most electrolytic capacitors are rated on the basis of minimum guaranteed value and the real value may be higher. ●

Over-Voltage Disconnect Switch

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Circuit designers often test prototypes with a variable-voltage bench power supply. Any accidental change in the voltage level could damage components in the prototype. This can be prevented by using an over-voltage disconnect circuit in series with the prototype under test.

The circuit is built around adjustable, precision shunt regulator TL431 (IC1), n-channel MOSFET IRFZ44 (T1) and a few discrete components. The heart of the circuit is the TL431 IC. The TL431 is a well-known voltage reference integrated circuit combining

reference voltage and an open-collector amplifier. It offers such advantages as simplicity of implementation and low cost.

The reference voltage at the refer-

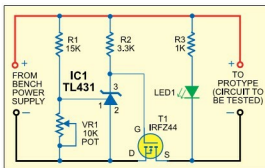


Fig. 1: Circuit of over-voltage disconnect switch

ence pin (pin 1) of shunt regulator IC1 is set by the divider network comprising resistor R1 and a 10-kilo-ohm preset. Here the set reference voltage is compared with the internal reference voltage of regulator IC1. As long as the set reference voltage at pin 1 is below the internal reference voltage, the IC allows the MOSFET (T1) to conduct. On the other hand, when the set reference voltage at pin 1 is above the



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